

Syllabus

Learning and Behavior (PSY/NSC 3320)

Jean P. Bodet III, Ph.D.

Baylor University

First half of the semester:

Behaviorism and Learning/Memory

Second half of the semester:

Cognitive Neuroscience of Learning/Memory

Introduction & Methods

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Learning and Behavior

The research that led to the modern-day
psychological approaches to studying
learning via behavior
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Behaviorism

“Purely objective experimental branch of natural
science” (Watson, 1913)

“Approach to psychology that emphasizes the study of
environmental influences on observable behavior”

Our book

Learning and Behavior

Behaviorism: studying *learning* by observing changes in *behavior* in response to some change in *environment*

Learning and Behavior

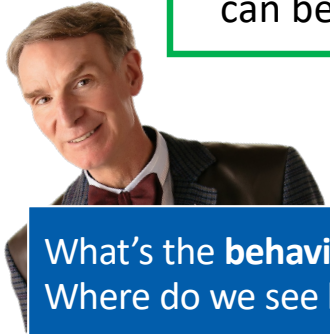
Learning: “relatively enduring change in behavior that results from some type of experience”

Behavior: “any activity of an organism that can be observed or somehow measured”

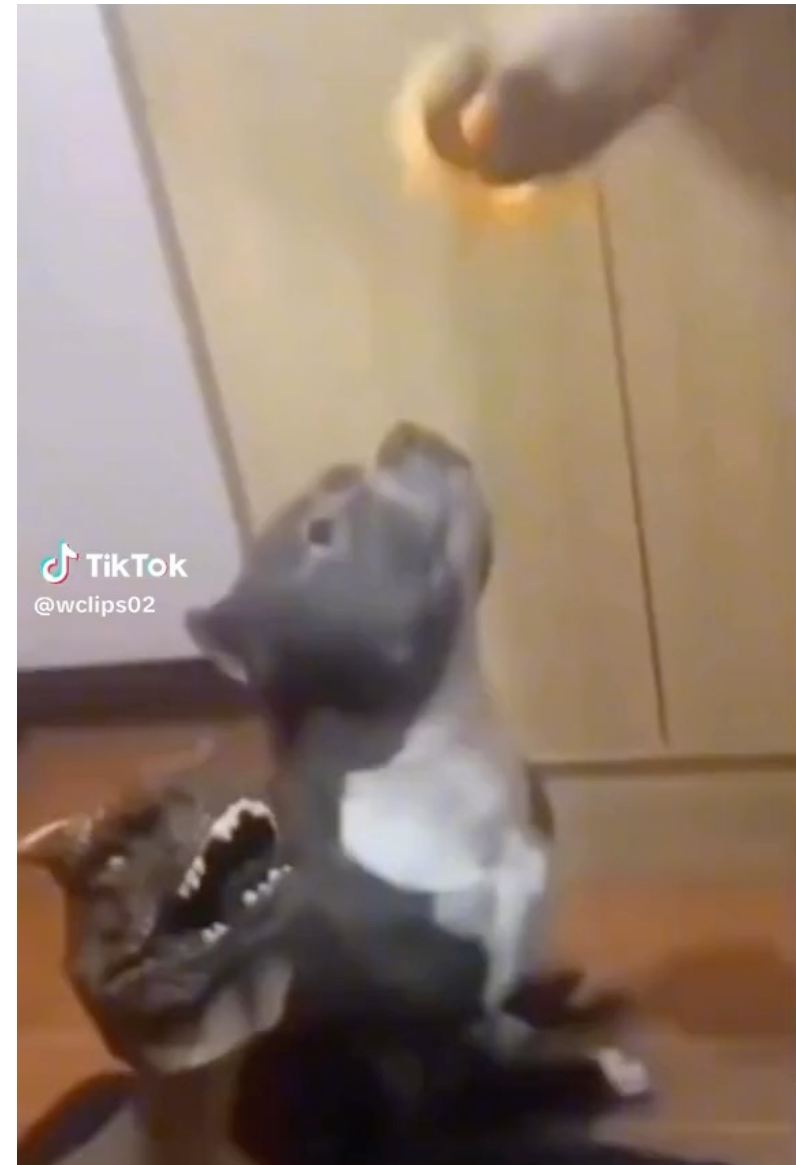
Learning and Behavior

Learning: “relatively enduring change in behavior that results from some type of experience”

Behavior: “any activity of an organism that can be observed or somehow measured”



What's the **behavior**?
Where do we see **learning**?



Learning and Behavior

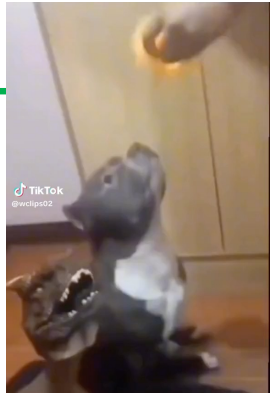
*Sometimes, the change in behavior resulting from learning is that the behavior becomes more solidified (the behavior **doesn't** increase/decrease)*

*Punish dino puppet in front of dog again
Dog **continues** to not take ball*

How does environment affect behaviors (how do we *learn* from experience)?
Five different 'schools of thought' on what's involved in this behaviorist question...

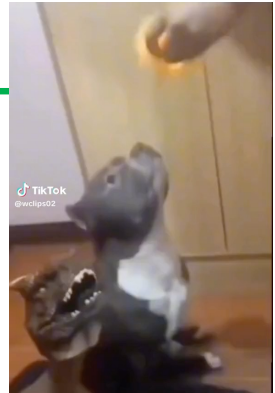
The Five Schools of Behaviorism:

- **Methodological Behaviorism**
- **Neobehaviorism**
- **Cognitive Behaviorism**
- **Social Learning Theory**
- **Radical Behaviorism**



After a stimulus, a given behavior changes...

- **Methodological Behaviorism** — *because of environment alone*
- **Neobehaviorism** — *when internal primary or secondary drives are satiated*
- **Cognitive Behaviorism** — *when internal expectations are met*
- **Social Learning Theory** — *because of a mix of environment + drives, expectations, etc.*
- **Radical Behaviorism** — *because of environment alone, though internal factors do exist*



After a stimulus, a given behavior changes...

- **Methodological Behaviorism** — *because of environment alone*

When responding to the ball, after observing the punishment, the biting stopped

- **Neobehaviorism** — *when internal primary or secondary drives are satiated*

When responding to the ball, after observing the punishment, the biting stopped because it satiated the drive to avoid pain

- **Cognitive Behaviorism** — *when internal expectations are met*

When responding to the ball, after observing the punishment, the biting stopped because it aligned with the dog's expectation that it would not be punished if it did not bite

- **Social Learning Theory** — *because of a mix of environment + drives, expectations, etc.*

When responding to the ball, after observing the punishment, the biting stopped based on any number of internal factors and their influence on the perception of the punishment

- **Radical Behaviorism** — *because of environment alone, though internal factors do exist*

When responding to the ball, after observing the punishment, the biting stopped (also, internal stuff exists but ignore those lol)

After a stimulus, a given behavior changes...

- **Methodological Behaviorism** — *because of environment alone*

When responding to the ball, after observing the punishment, the biting stopped

- **Neobehaviorism** — *when internal primary or secondary drives are satiated*

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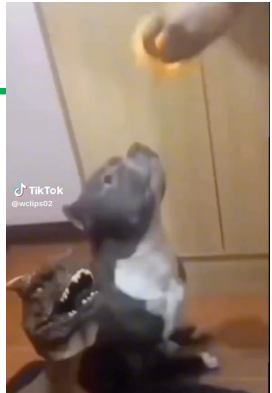
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- **Radical Behaviorism** — *because of environment alone, though internal factors do exist*

When responding to the ball, after observing the punishment, the biting stopped (also, internal mechanisms exist lol)

Certain **methods** are consistent in behaviorism research, regardless how you explain what the research finds...



Methods in Behaviorism

Stimulus-Response (S-R)

- Stimulus = **IV** = elicits behavior
- Response = **DV** = behavior of interest

Methods in Behaviorism

Stimulus-Response (S-R)

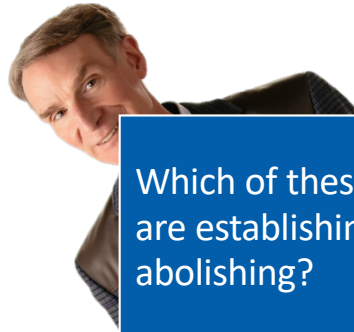
- Stimulus = **IV** = elicits behavior
 - Appetitive stimuli – event an organism seeks out (food)
 - Aversive stimuli – event an organism avoids (pain)
- Response = **DV** = behavior of interest

Methods in Behaviorism

Stimulus-Response (S-R)

- Stimulus = **IV** = elicits behavior
 - Appetitive stimuli – event an organism seeks out (food)
 - Aversive stimuli – event an organism avoids (pain)
 - Manipulating the value of a stimulus...
 - Establishing Operation – increase appetitiveness/aversiveness of event – *usually* deprivation
 - Abolishing Operation – decreases appetitiveness/aversiveness of event – *usually* satiation
- Response = **DV** = behavior of interest

A child hates broccoli and their parent forces them to eat it when they misbehave

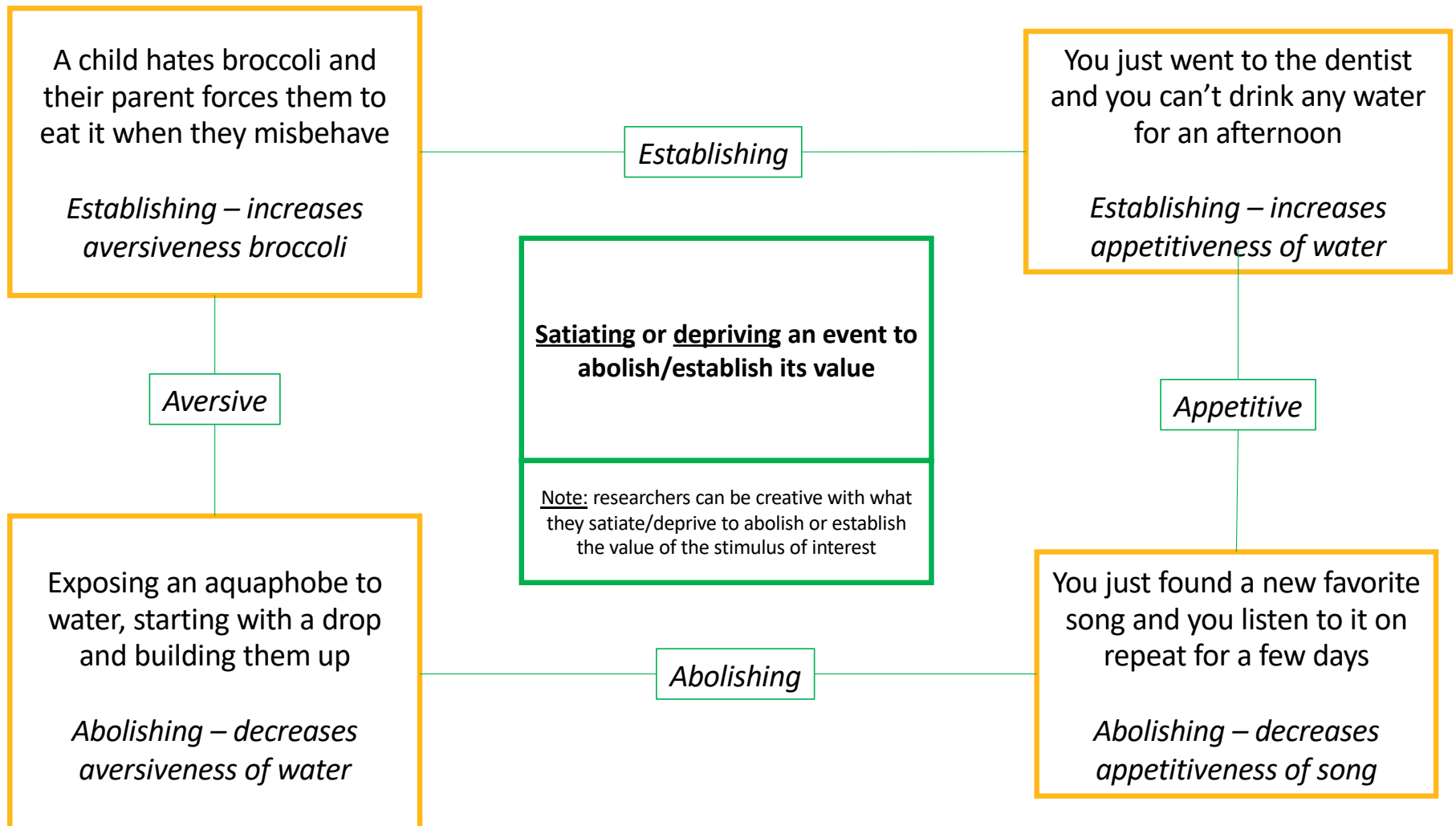


Which of these operations are establishing and which abolishing?

You just went to the dentist and you can't drink any water for an afternoon

Exposing an aquaphobe to water, starting with a drop and building them up

You just found a new favorite song and you listen to it on repeat for a few days



Methods in Behaviorism

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Manipulating Stimuli:

- Simple-Comparison (A-B)
- Reversal (A-B-A-B)
- Multiple-Baseline
- Changing-Criterion

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Goal: establish that changes in the IV (Stimulus) result in corresponding changes in the DV (Response)

*AKA – establish a behavioral **Functional Relationship***

Methods in Behaviorism

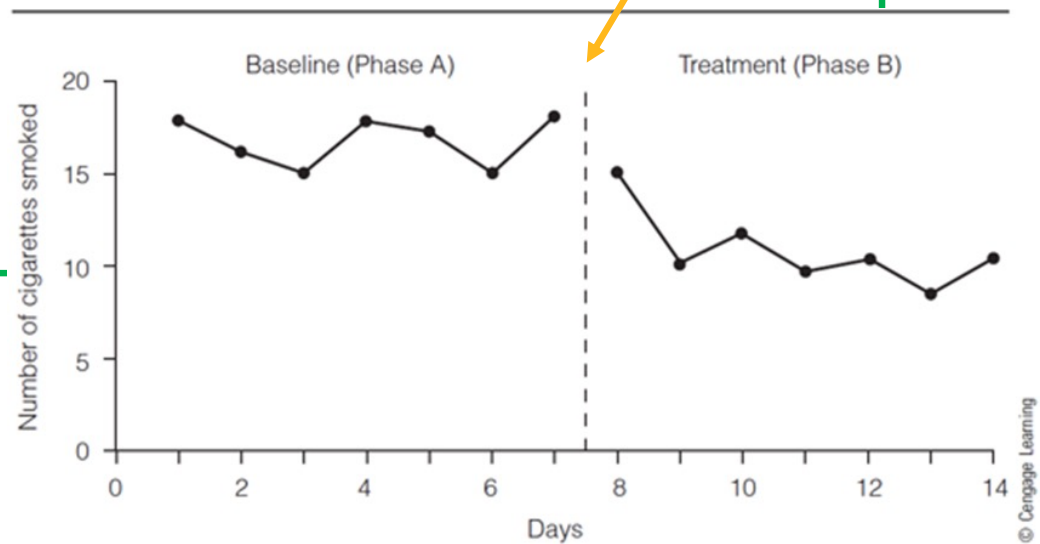
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Manipulating Stimuli:

- **Simple-Comparison (A-B)**
- Reversal (A-B-A-B)
- Multiple-Baseline
- Changing-Criterion

1. Gather baseline data
2. Introduce stimulus
3. Observe change in data



What may be the concern here...?

Methods in Behaviorism

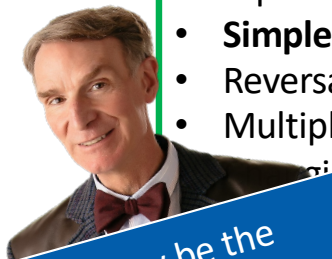
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Manipulating Stimuli:

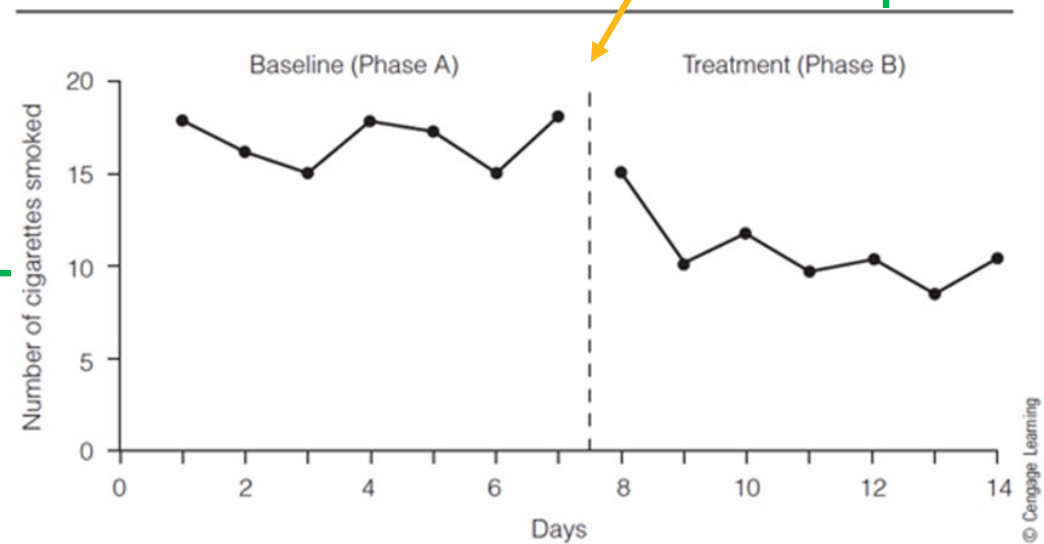
- **Simple-Comparison (A-B)**
- Reversal (A-B-A-B)
- Multiple-Baseline
- Meeting-Criterion

1. Gather baseline data
2. Introduce stimulus
3. Observe change in data



What may be the concern here...?

How do we make sure something else didn't cause the change?



Methods in Behaviorism

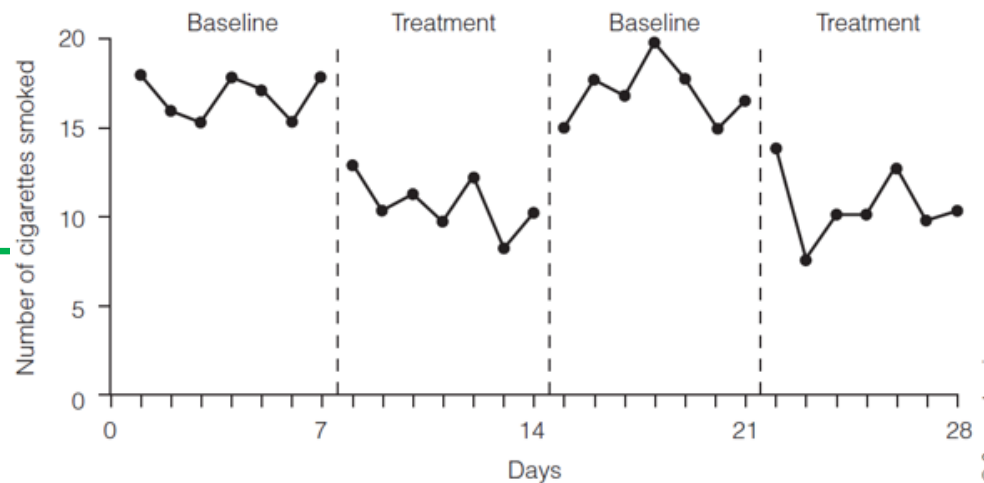
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Manipulating Stimuli:

- Simple-Comparison (A-B)
- **Reversal (A-B-A-B)**
- Multiple-Baseline
- Changing-Criterion

Introduce/remove stimulus repeatedly
Control for possibility of other effects



What might go wrong here?

Methods in Behaviorism

Stimulus-Response (S-R)

- Stimulus = **IV** = elicits behavior
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Introduce/remove stimulus repeatedly
Control for possibility of other effects

Manipulating Stimuli:

Comparison (A-B)

(A-B)

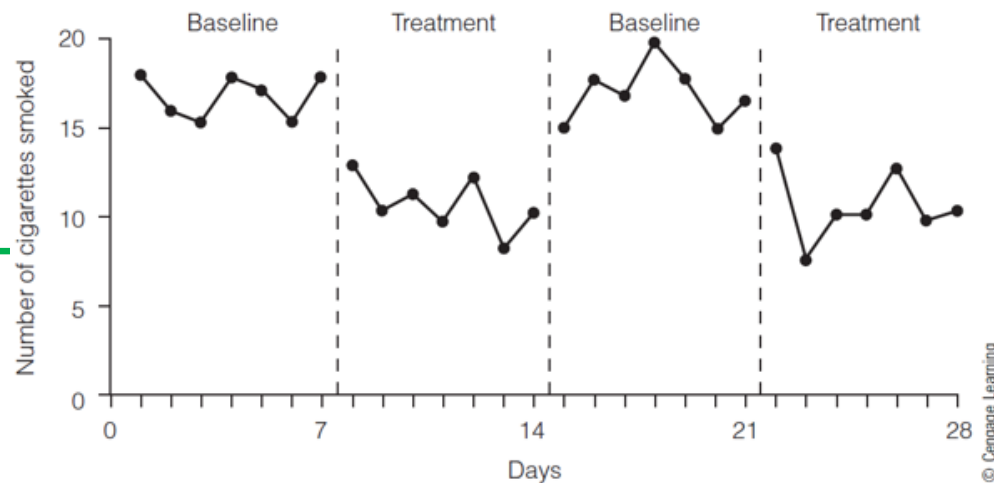
line

orion

What might go wrong here?

What if data doesn't return to baseline in 2nd baseline period?

...only appropriate if stimulus effect is impermanent



Methods in

Manipulate *when* stimulus is introduced *across* subjects

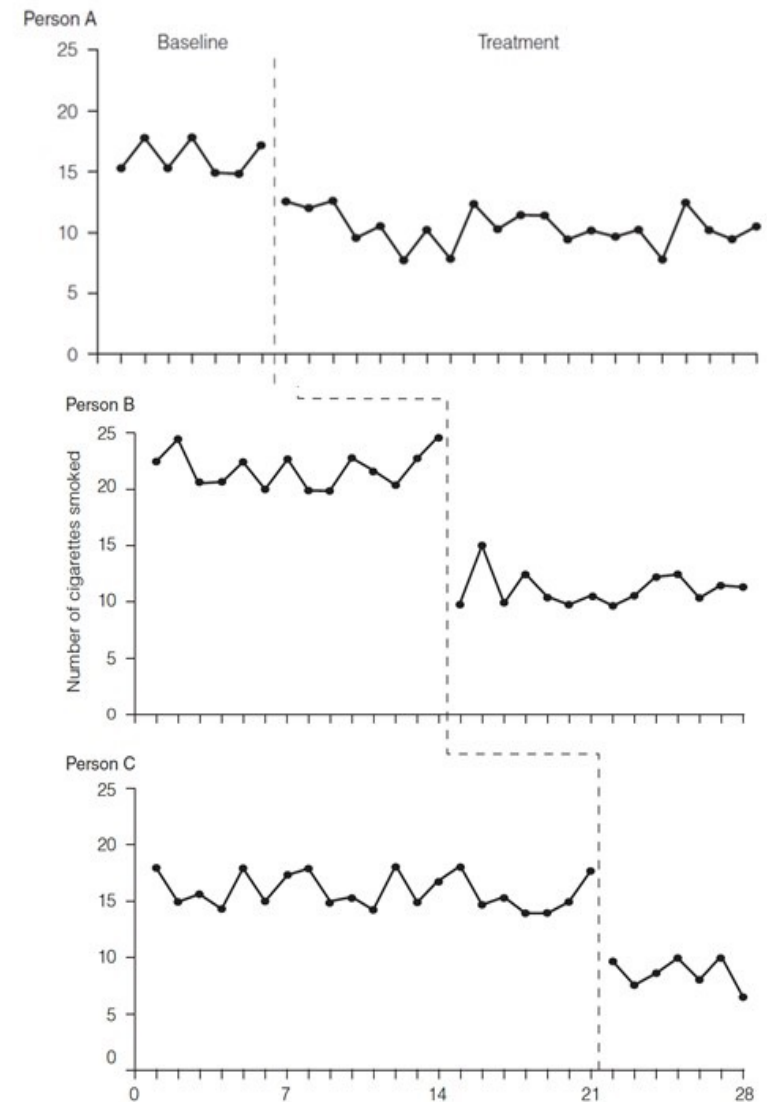
Useful when stimulus effect is permanent (so, can't use Reversal) & want to make sure something else didn't cause the change (which Simple Comparison can't prove)

Stimulus-Response (S-R)

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Manipulating Stimuli:

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- Reversal (A-B-A-B)
- **Multiple-Baseline**
- Changing-Criterion



Methods in Behaviorism

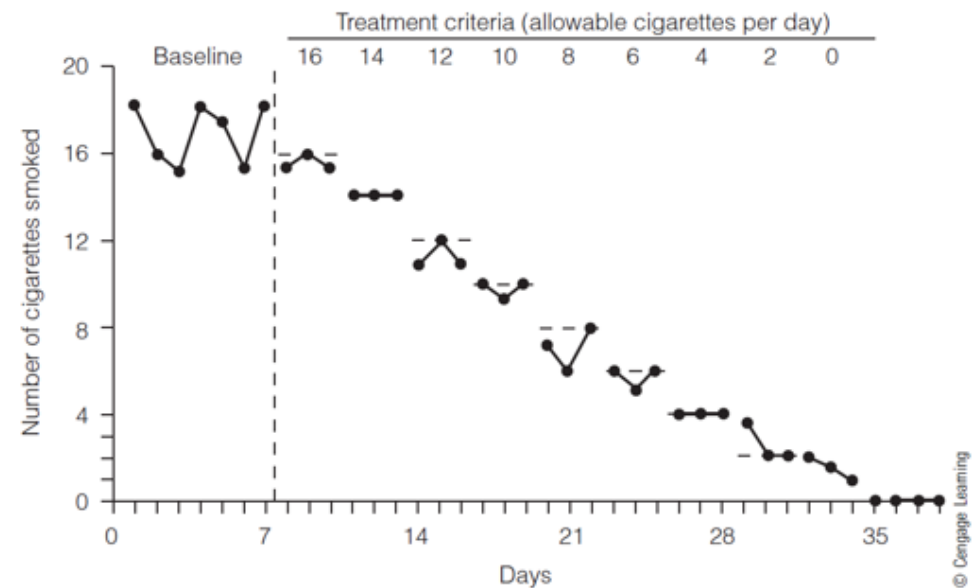
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- Response = **DV** = behavior of interest

Manipulating Stimuli:

- Simple-Comparison (A-B)
- Reversal (A-B-A-B)
- Multiple-Baseline
- **Changing-Criterion**

- Check effect of gradual change of stimulus
- Changes in DV must *really* match changes in IV to suggest functional relationship



Behaviors we might measure...

- First – target behavior must be observable & unambiguous
- **Rate of response** – count target behaviors in period of time (lever presses/minute)
- **Intensity** – magnitude of behavior (amount of spit)
- **Duration** – length of time of behavior, regardless rate (3 minutes of lever presses)
- **Speed** – how long it takes to complete behavior (rat took 2 minutes to escape maze)
- **Latency** – time before behavior begins (lever pressing began 3 minutes after stimulus)
- **Interval recording** – number of time periods in consecutive recording where behavior occurred (60 minutes split into 10-minute intervals, 4 contained target behavior)
- **Time-Sample** – number of time periods in nonconsecutive recording where behavior occurred (3 days, 6 random 10-minute intervals, 4 contained target behavior)
- **Topography** – physical manifestation of behavior (which hand performed target behavior; exactly how is child being disruptive)
- **Error counts** – count unsuccessful attempts at target behavior before successful attempt

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How to confirm behavioral observations are accurate?

Multiple observers – check **interobserver reliability**

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- **Topography**
- **Error**

scored-interval IOA

interval #	1	2	3	4	5	6	7	8	9	10
observer 1	+	+	+	-	+	-	-	+	-	-
observer 2	-	+	+	-	-	-	+	-	-	+

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Interobserver reliability:

Number of times that observers agree on what they observed per period of time (shoot for 80%)

Behavioral measure...

- First – target behavior must be clearly unambiguous
- **Rate of response** – count target behavior over a period of time (lever presses/minute)
- **Intensity** – magnitude of behavior (e.g., number of lever presses)
- **Duration** – length of time behavior lasts (e.g., minutes to escape maze)
- **Speed** – how long it takes to complete a task (e.g., minutes after stimulus)
- **Latency** – time between stimulus and response (e.g., minutes after stimulus)
- **Interval recording** – number of times behavior occurred during a period of time (e.g., minutes after stimulus)
- **Time-Sample** – number of time periods in nonconsecutive recording where behavior occurred (3 days, 6 random 10-minute intervals, 4 contained target behavior)
- **Topography** – description of the form of the behavior
- **Error** – number of times behavior was incorrectly recorded

What's the inter-observer reliability here? Are these observations sufficiently reliable?

scored-interval IOA

interval #	1	2	3	4	5	6	7	8	9	10
observer 1	+	+	+	-	+	-	-	+	-	-
observer 2	-	+	+	-	-	-	+	-	-	+

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Interobserver reliability:

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Questions?